

Continuous functions – 1

Rigidity: continuity, density of the rationals, pathologies

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Theoretical minimum.

Def. f is continuous at x_0 if for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - f(x_0)| < \varepsilon$ whenever $|x - x_0| < \delta$; equivalently, $x_n \rightarrow x_0$ implies $f(x_n) \rightarrow f(x_0)$. The *oscillation* of f at x_0 is $\text{osc } f(x_0) = \lim_{\delta \rightarrow 0} (\sup_{|x-x_0|<\delta} f - \inf_{|x-x_0|<\delta} f)$, and f is continuous at x_0 exactly when this number is 0.

Theorem 1 (values on a dense set). Two continuous functions agreeing on a dense set coincide. Hence a continuous function is determined by its restriction to $\mathbb{Q}$, and every constraint imposed on the rationals propagates to the whole line — this is the mechanism behind almost every problem of the sheet.

Theorem 2 (intermediate value property). A continuous function on an interval takes every intermediate value; the image of an interval is an interval, and the image of a segment is a segment.

Fact 1. A monotone function has one-sided limits at every point, its set of discontinuities is at most countable, and every jump is a genuine gap in the range. Continuity plus injectivity on an interval already forces strict monotonicity.

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1. (IMC 2005) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $(f(x))^n$ is a polynomial for every $n = 2, 3, \dots$. Does it follow that f is a polynomial?
 2. (IMC 2008) Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) - f(y)$ is rational for all reals x, y with $x - y$ rational.
 3. (MIPT 1993) Is there a continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$ taking rational values at irrational points and irrational values at rational ones?

4. (IMC 2008) Suppose f and g are real-valued functions on the line with $f(r) < g(r)$ for every rational r . Does this imply $f(x) \leq g(x)$ for every real x if
- a) f and g are non-decreasing;
 - b) f and g are continuous?
5. (MIPT 2009) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be such that at every point x_0 the finite limit $g(x_0) = \lim_{x \rightarrow x_0} f(x)$ exists. Prove that g is continuous.